

 **TARGET CAT 2025** 



MBA RAPID FIRE

QUANT : Algebra

Algebra **Part II**

Lecture No.- 02

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Recap of Previous Lecture :

- Linear Equation
- Exponents
- Quadratic Equations

TOPICS *to be covered*



- 1) Revision & Numerical based on 'Inequalities'
- 2) Revision & Numerical based on 'Logarithm'
- 3) Revision & Numerical based on 'Function'

QUESTION-1

8

$$a=1, b=6, c=-3$$

$a, b \rightarrow \text{integer}$



#Q. If the minimum value of the quadratic expression $x^2 + ax + 12 = 0$ is equal to the minimum value of the quadratic expression $x^2 + bx - 3 = 0$, how many integer pairs (a, b) satisfy this condition?

odd + odd = even
even + even = even

$$x^2 + ax + 12 = 0$$

$a > 0$

$$\text{Min Value} = \frac{-D}{4a} = \frac{-[b^2 - 4ac]}{4a}$$

$$(16, -14)$$

$$(16, 14)$$

$$= \frac{4ac - b^2}{4a}$$

$$(a+b)(a-b) = 60$$

$$2 \times 30 = 60$$

$$30 \times 2 = 60$$

$$6 \times 10 = 60$$

$$10 \times 6 = 60$$

$$(-2) \times (-30) = 60$$

$$(-30) \times (-2) = 60$$

$$(-6) \times (-10) = 60$$

$$(-10) \times (-6) = 60$$

$$\frac{4(1)(12) - a^2}{4(1)} = \frac{4(1)(-3) - b^2}{4(1)}$$

$$48 - a^2 = -12 - b^2$$

$$60 = a^2 - b^2$$

$$a^2 - b^2 = 60$$

$$(a+b)(a-b) = 60$$

$$a + b = 2$$

$$a - b = 30$$

$$2a = 32$$

$$a = 16$$

$$b = -14$$

$$(16, -14)$$

$$1 \times 60 = 60$$

$$2 \times 30 = 60$$

$$3 \times 20 = 60$$

$$4 \times 15 = 60$$

$$5 \times 12 = 60$$

$$6 \times 10 = 60$$

$$(-1) \times (-60) = 60$$

$$(-2) \times (-30) = 60$$

$$(-3) \times (-20) = 60$$

$$(-4) \times (-15) = 60$$

$$(-5) \times (-12) = 60$$

$$(-6) \times (-10) = 60$$

Inequalities



Inequalities:



$$3x + 8 > 11$$

$$3x > 3$$

$$x > 1$$

$$(1, \infty)$$



$\leq$
 $\geq$

$() \rightarrow \text{open}$

$[] \rightarrow \text{closed}$

$$x \geq 1$$



$$[1, \infty)$$



$$(2, 5]$$



Inequalities :



Points to Remember :

$$\begin{array}{ccc} > & \rightarrow & < \\ < & \rightarrow & > \\ \geq & \rightarrow & \leq \\ \leq & \rightarrow & \geq \end{array}$$

$$\begin{array}{l} \frac{3x}{4} \geq 12 \\ 3x \geq 48 \\ x \geq 16 \end{array}$$

$$\begin{array}{l} \frac{3x}{-4} \geq 12 \\ \textcircled{-4} \\ \textcircled{3}x \leq -48 \\ x \leq -16 \end{array}$$

1. Many simple inequalities can be solved by adding, subtracting, multiplying or dividing both sides

2. But, Following things will change direction of the inequality:

- Multiplying or dividing both sides by a **negative** number
- Swapping left and right hand sides

$$\begin{array}{l} a = b \\ b = a \end{array} \quad \begin{array}{l} a < b \\ b > a \end{array}$$

3. Don't multiply or divide by a **variable** (unless you know it is always positive or always negative).

$$\begin{array}{l} \boxed{3 \geq 8x} \\ \frac{3}{x} \geq 8 \end{array} \quad / \quad \begin{array}{l} \frac{3}{\textcircled{-5}} \geq 8 \\ 3 \leq 8x \end{array}$$



Wavy Curve Method :



$$\underbrace{(x-a)}_{x=a} \cdot \underbrace{(x-b)}_{x=b} \cdot \underbrace{(x-c)}_{x=c} \geq 0$$



$$[a, b] \cup [c, \infty)$$



Wavy Curve Method :



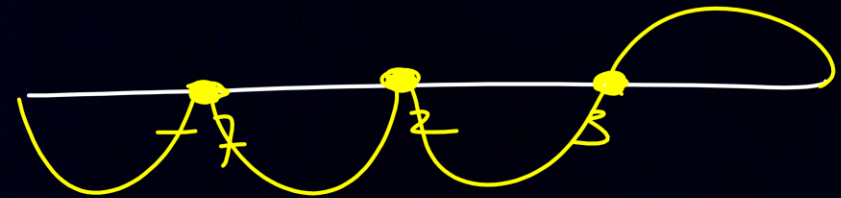
$$(\underline{x-3})^{\textcircled{3}} \cdot (\underline{x-5})^{\textcircled{1}} \cdot (\underline{x-6})^{\textcircled{7}} < 0$$



$$(-\infty, 3) \cup (5, 6)$$

$$(\underline{x-2})^{\textcircled{4}} \cdot (\underline{x-3})^{\textcircled{5}} \cdot (\underline{x+7})^{\textcircled{8}} \leq 0$$

$x=2 \quad x=3 \quad x=-7$



$$(-\infty, 3]$$

QUESTION- 2



#Q. Find the number of integer solutions for the inequality :

$$(x^2 - 7x + 10)(x^2 - 12x + 35) \leq 0$$

$$(\underline{x-5})(x-2)(x-7)(\underline{x-5}) \leq 0$$

$$(\underline{x-2})(\underline{x-5})^2(\underline{x-7}) \leq 0$$

$\underline{x=2} \quad \underline{x=5} \quad \underline{x=7}$

A

4



B

6

$$x \rightarrow [2, 7]$$

C

8

$$2, 3, 4, 5, 6, 7$$

D

10

$$x^2 - 7x + 10$$

$$\boxed{a=1}, b=-7, c=10$$

$$x \rightarrow (10)$$

$$+ \rightarrow -7$$

$$(-5)$$

$$(-2)$$

$$x = \frac{-5}{-1} = 5$$

$$x = \frac{-2}{-1} = 2$$

$$(x-5)$$

$$(x-2)$$

$$x^2 - 12x + 35$$

$$x \rightarrow 35$$

$$+ \rightarrow -12$$

$$-7$$

$$-5$$

$$(x-7)(x-5)$$

QUESTION- 3



#Q. Find the range of x such that

$$\frac{(x^2-9)(x-5)}{(x^2-4x+4)} \geq 0$$

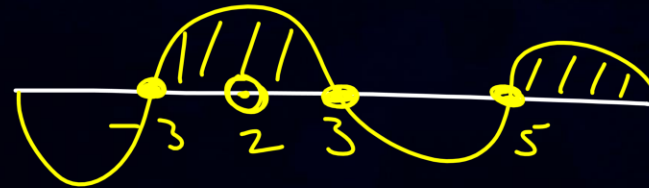
$$\frac{(x-3)(x+3)(x-5)}{(x-2)^2} \geq 0$$

$$x-2=0$$

$$x=2$$

$$(x-3)(x+3)(x-5) \geq 0$$

$$x=-3$$



OR

$$[-3, 2) \cup (2, 3] \cup [5, \infty)$$

$$[-3, 3] \cup [5, \infty) - \{2\}$$

A $[-\infty, 3] \cup [5, \infty)$

B $[-3, 0] \cup [5, \infty)$

C $[-3, 3] \cup [5, \infty)$

D $[-3, 2) \cup (2, 3] \cup [5, \infty)$



Solving Higher Order Polynomial :



$$ax^3 + bx^2 + cx + d = 0$$

$$1x^3 - 6x^2 + 11x - 6 = 0$$

$$x = 1, 2, 3$$

$$2 \rightarrow x = 1$$

$$1^3 - 6(1) + 11(1) - 6 = 0$$

$$12 - 12 = 0$$
$$0 = 0$$

$$\text{Sum of Root} = \left[\frac{-b}{a} \right] = -\frac{(-6)}{1} = 6$$

$$\text{Product of Root} = \left[\frac{-d}{a} \right] = -\frac{(-6)}{1} = 6$$

①	①	-6	11	-6
	↓			
	1	-5	6	0

$$(x^2 - 5x + 6) = 0$$

$$x \rightarrow 6$$
$$+ \rightarrow -5$$

$$x = 3$$

$$x = 2$$

$$\text{(1)} \quad 2 + \beta + \gamma = 6$$

$$\beta + \gamma = 5$$

$$\text{(2)} \quad 2 \cdot \beta \cdot \gamma = 6$$

$$\beta \cdot \gamma = 6$$

$$\beta = 2$$
$$\gamma = 3$$

QUESTION- 4



#Q. The smallest integer n such that $(n^3 - 11n^2 + 32n - 28) > 0$ is [CAT 2018]

$$n^3 - 11n^2 + 32n - 28 = 0$$

$n \neq 1$

$$1 - 11(1) + 32(1) - 28$$

$n = 2$

$$8 - 11(4) + 32(2) - 28 = 0$$

$$72 - 72 = 0$$

$$0 = 0$$

$(n - 2) = 0$

$$\begin{array}{c|ccc} 2 & 1 & -11 & 32 & -28 \\ & \downarrow & & & \\ & 1 & -9 & 14 & 0 \end{array}$$

$$x - 9x + 14 = 0$$

$$\begin{array}{l} x \rightarrow 14 \\ + \rightarrow -9 \end{array} < \begin{array}{l} -7 \\ -2 \end{array}$$

$$\begin{array}{l} n = 2 \Rightarrow (n - 2) \\ n = 7 \Rightarrow (n - 7) \end{array}$$

$$(n - 2)^2 (n - 7) > 0$$



QUESTION- 5



#Q. If x is real, then the sum of maximum and minimum values of the expression

$$\frac{x^2 - 3x + 4}{x^2 + 3x + 4} \text{ will be}$$

$$y = \frac{x^2 - 3x + 4}{x^2 + 3x + 4}$$

$$y_{\max} + y_{\min} = ?$$

$$\frac{7}{1} + \frac{1}{7} = \frac{50}{7}$$

$$D \geq 0$$

$$y^2 - 50y + 7$$

$$x \rightarrow 49$$

$$+ \rightarrow -50$$

$$\begin{pmatrix} -49 \\ -1 \end{pmatrix}$$

$$y x^2 + 3yx + 4y = x^2 - 3x + 4$$

$$(y x^2 - x^2) + \underline{3yx + 3x} + 4y - 4 = 0$$

$$x^2(y - 1) + 3x(y + 1) + 4(y - 1) = 0$$

$$a = y - 1 \quad b = \underline{3(y + 1)} \quad c = 4(y - 1)$$

$$b^2 - 4ac \geq 0$$

$$9(y + 1)^2 - 4(y - 1)4(y - 1) \geq 0$$

$$9(y^2 + 2y + 1) - 16(y^2 - 2y + 1) \geq 0$$

$$9y^2 + 18y + 9 - 16y^2 + 32y - 16 \geq 0$$

$$-7y^2 + 50y - 7 \geq 0$$

$$7y^2 - 50y + 7 \leq 0$$

$$(y - 7) \cdot (y - \frac{1}{7}) \leq 0$$



$$y \rightarrow [\frac{1}{7}, 7]$$

A $y = \frac{-49}{-7} = 7$
50/7 (y - 7)

B 49/7
 $y = \frac{-1}{-7} = \frac{1}{7}$
(y - 1/7)

C 7/49

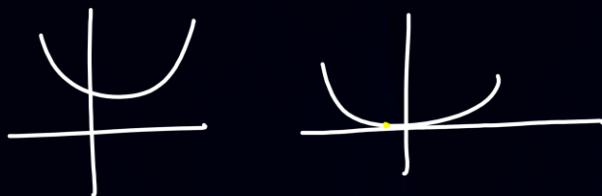
D 7/50

QUESTION- 6



#Q. How many integer values of k satisfy the inequality $\underline{x^2 - 6kx + 5k^2 + 16} > 0$

$y > 0$ $y \geq 0$



$a = 1$, $b = -6k$, $c = 5k^2 + 16$

$D < 0 \rightarrow \text{Imaginary}$

$b^2 - 4ac < 0$

$(-6k)^2 - 4(1)(5k^2 + 16) < 0$

$36k^2 - 20k^2 - 64 < 0$

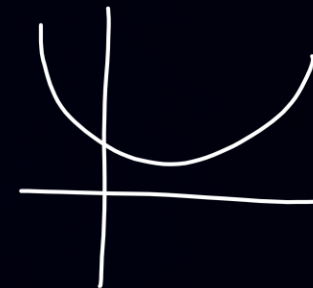
$16k^2 - 64 < 0$

$16k^2 < 64$

$k^2 < 4$

$k^2 - 2^2 < 0$

$(k+2)(k-2) < 0$



A 1

B 3



$k \rightarrow (-2, 2)$

$k \rightarrow -1, 0, 1$

C 5

D Infinite

Logarithm



Logarithmic Functions :



$\log_m a \rightarrow$ log of a to the base m .

$a \rightarrow$ argument

$m \rightarrow$ base

$$a, m > 0$$

$$m \neq 1$$



Logarithmic Functions :



Important Properties:

1. $\log_a b = \frac{\log b}{\log a}$

2. $\log_a b = \frac{1}{\log_b a}$

3. $\log_a m + \log_a n = \log_a mn$

4. $\log_a m - \log_a n = \log_a \frac{m}{n}$

5. $\log_a 1 = 0$

6. $\log_a a = 1$

7. $\log_b (a^m) = \underline{m} \cdot \log_b a$

8. $\log_{N^k} a = \left(\frac{1}{k}\right) \log_N a$

9. Conversion of Logarithmic form into Exponential form:
 $\log_a b = x \quad \Leftrightarrow \quad b = a^x$

10. Conversion of Exponential form into Logarithmic form:
 $p = q^r \quad \Leftrightarrow \quad \log_q p = r$

11. $a^{\log_a b} = b$

$\log_2 32 = 5$
 $32 = 2^5$

$\log(100) = 2$
 $\log(10) = 1$
 $\log(1) = 0$

QUESTION- 7



#Q. $(\log_4 5)^{\left[\frac{3}{\log_4 (\log_4 5)} \right]} = ?$

$\log_4 5 \rightarrow x$

$(\log_4 5)^{3 \cdot \log_{(4)} (4)}$

$(\log_4 5)^{[\log_{(4)} 5] \cdot 3}$

$(\cancel{x})^{\cancel{3} \cdot 3} = 64$

$\log_b a = \frac{1}{\log_a b}$

$\log(a^m) = m \cdot \log a$

$$n \neq 1$$

3

$$n \geq 1$$

$$\log_2 n \rightarrow +ve$$

$$n \rightarrow (-1, 4]$$

$$n \rightarrow \cancel{X}, \boxed{2, 3, 4}$$

$$\log_m 9 \geq \log_m 6 \quad \dots m > 1$$

$$a \geq 6$$

$$\log_m a \geq \log_m b \quad (0 < m < 1)$$

$$a \leq b$$

$X \rightarrow -5$
 $t \rightarrow -3$

-5 1

$$\log_2(3n+4) \geq \log_2 n^2$$

$$(3n+4) \geq n^2$$

$$(3n+4) > n^2$$

$$0 > n^2 - 3n - 4$$

$$(n-4)(n+1) \leq 0$$

$$\underline{\underline{(n-4)}} \quad \underline{\underline{(n+1)}} \leq 0$$

QUESTION- 9

$$ax^3 + bx^2 + cx + d = 0$$



#Q. Find the sum of all real values of x for which $\frac{\log(x^2 - 2) + \log(x + 6)}{\log(2x)} = 2$.

2	1	2	-2	-12
	↓	2	8	12
	①	4	6	8

A 2

B -2

C 6

D -6

$$x^2 + 4x + 6 = 0$$

$$a=1, b=4, c=6$$

$$b^2 - 4ac = 16 - 4(1)(6) = -8 < 0$$

Both roots are imaginary

$$\log[(x^2 - 2)(x + 6)] = 2 \log(2x)$$

$$\log[x^3 + 6x^2 - 2x - 12] = \log(4x^2)$$

$$x^3 + 6x^2 - 2x - 12 = 0 \Rightarrow \text{Sum of Roots}$$

$$x = 2 \quad (\text{Real})$$

$$8 + 2(4) - 2(2) - 12 = 0$$

$$16 - 16 = 0$$

$$0 = 0$$

$$= \frac{-b}{a} = \frac{-2}{1} = -2$$

QUESTION- 10



#Q. If $\log 7 + \left[1 + \frac{1}{2^x}\right] \log 2 = \log [\sqrt[x]{2} - 32]$, then find the value of $24x$.

Youtube

[Arithmetic & Algebra]
Mixed Bag

Q. No. 1

↓
TRY

[27th Oct 2025]

A 2

B -2

C 6

D -6



Functions



Basic Function :

$$f(x) = 3x$$
$$g(x) = 5x + 8$$



$$f(x) = 3x + 5$$

$$x = 2 \quad f(2) = 3(2) + 5 = 11$$

$$x = -1 \quad f(-1) = 3(-1) + 5 = 2$$

$$x = 0 \quad f(0) = 3(0) + 5 = 5$$

$$f(x) \pm g(x)$$

$$\begin{array}{r|l} = 3x + (5x + 8) & 3x - (5x + 8) \\ = 8x + 8 & -2x - 8 \end{array}$$

$$\begin{array}{r|l} \underline{f(x)} \cdot \underline{g(x)} & \frac{f(x)}{g(x)} = \frac{3x}{5x+8} \\ 3x(5x+8) & \\ 15x^2 + 24x & \end{array}$$



Basic Function :



Composite function:

$$\boxed{f(\underline{x}) = (3x + 4)} \quad \underline{g(\underline{x})} = (x - 7)$$

$$\begin{aligned} f[g(\underline{x})] &= 3[g(x)] + 4 \\ &= 3[x - 7] + 4 \\ &= 3x - 21 + 4 \\ &= \boxed{3x - 17} \end{aligned}$$

$$\begin{aligned} g[f(\underline{x})] &= f(x) - 7 \\ &= 3x + 4 - 7 \\ &= \underline{\underline{3x - 3}} \end{aligned}$$

QUESTION- 11

$$A = B$$

$$B = A$$



#Q. If $\underline{(x+1)} + f(x-1) = f(x)$ is defined for all real 'x' and $f(3) = 10$, find the value of $f(49)$.

$$\underline{x+1} + f(x-1) = f(x)$$

$$(x+1) = f(x) - f(x-1)$$

$$\begin{aligned} & \text{①} + \text{②} + \text{③} + \text{④} + 5 + 6 + \dots + 50 \\ &= \frac{n(n+1)}{2} = \frac{50(51)}{2} = 1275 \end{aligned}$$

$$\therefore 5 + 6 + 7 + \dots + 50 = 1275 - 10 = 1265$$

$$f(x) - f(x-1) = x+1$$

$$\cancel{f(4)} - \underline{f(3)} = 5$$

$$\longrightarrow x = 4$$

$$+ \cancel{f(5)} - \cancel{f(4)} = 6$$

$$\longrightarrow x = 5$$

$$+ \cancel{f(6)} - \cancel{f(5)} = 7$$

$$\longrightarrow x = 6$$

$$+ \cancel{f(7)} - \cancel{f(6)} = 8$$

$$\longrightarrow x = 7$$

+
+
+
+
+

$$\underline{f(49)} - \cancel{f(48)} = 50$$

$$f(49) - f(3) = 5 + 6 + 7 + \dots + 50$$

$$\longrightarrow x = 49$$

$$f(49) - f(3) = 1265$$

$$f(49) = 1275$$

A

1265

B

1275

C

1285

D

1295

QUESTION- 12



#Q. For any non-zero real number x , let $f(x) + 2f\left(\frac{1}{x}\right) = 3x$. Then, the sum of all possible values of x for which $f(x) = 3$ is

[CAT 2024]

A 3

B -3

C 2

D -2

$$f(x) + 2f\left(\frac{1}{x}\right) = 3x$$

$$\text{put } x = \frac{1}{x}$$

$$f\left(\frac{1}{x}\right) + 2f(x) = 3 \cdot \frac{1}{x}$$

$$f\left(\frac{1}{x}\right) = \left(\frac{3}{x} - 2f(x)\right)$$

$$\rightarrow \text{eq}^1 \text{ (1)}$$

$$f(x) + 2\left[\frac{3}{x} - 2f(x)\right] = 3x$$

$$f(x) + \frac{6}{x} - 4f(x) = 3x$$

$$\frac{6}{x} - 3x = 3f(x)$$

$$\frac{6}{x} - 3x = 3(3)$$

$$\frac{6}{x} - 3x = 9$$

$$6 - 3x^2 = 9x$$

$$-3x^2 - 9x + 6 = 0$$

$$3x^2 + 9x - 6 = 0$$

$$x^2 + 3x - 2 = 0$$

$$\text{Sum} = \frac{-b}{a} = \frac{-3}{1} = -3$$



2 Mins Summary

- 1) Revision & Numerical based on 'Inequalities'
- 2) Revision & Numerical based on 'Logarithm'
- 3) Revision & Numerical based on 'Function'



*Thank
You*